

MagicLoader

Implementation Guide

for Windows[®] CE

MMSP[™] 2

User Manual

MDMP2520FMLUM-WC

Ver. 2.1 / November 2, 2005

MP2520F

*MagicLoader Implementation guide for Windows[®] CE.NET
for MMSP 2 Application Processor*

MAGIC EYES

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Document History

Version	History	Date
1.0	MagicLoader Implementation guide	2005. 9. 8
2.0	-Add file download function from PC to NAND flash thru USB -Add Logo graphic file display function	2005.10.27
2.1	-Add Make logo graphic file	2005.11.2

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1. MagicLoader

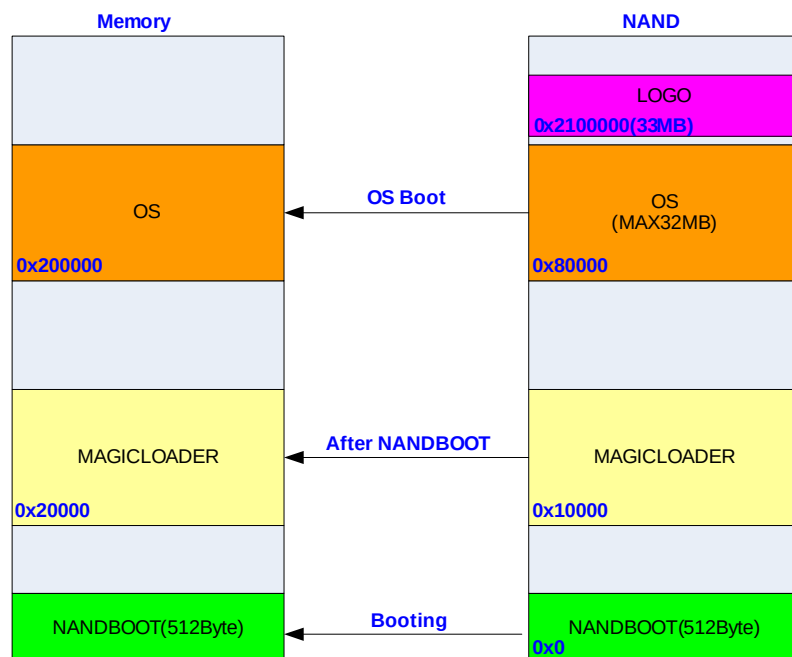
1.1. What is the MagicLoader

Generally, MP2520F needs two bootloaders. NANDBOOT.bin is the first bootloader for reading the second bootloader. MagicLoader.bin(or EBOOT.nb0) is the second bootloader for reading the OS image file. Boot sequences of MP2520F are as follows :

- (1) When reset signal is asserted to MP2520F, MP2520F copies the first bootloader(NANDBOOT.bin) from NAND flash to SDRAM address 0
- (2) NANDBOOT.bin is executed (MagicLoader.bin(or EBOOT.nb0) is copied from NAND flash to SDRAM address 0x20000)
- (3) MagicLoader.bin(or EBOOT.nb0) is executed (OS image file is copied from NAND flash to SDRAM address 0x200000)

(Caution)

- The maximum size of OS image file can not over 32MB
- The maximum size of Logo graphic file can not over 1MB



<Figure 1-1 >

1.2. The Functions of the MagicLoader

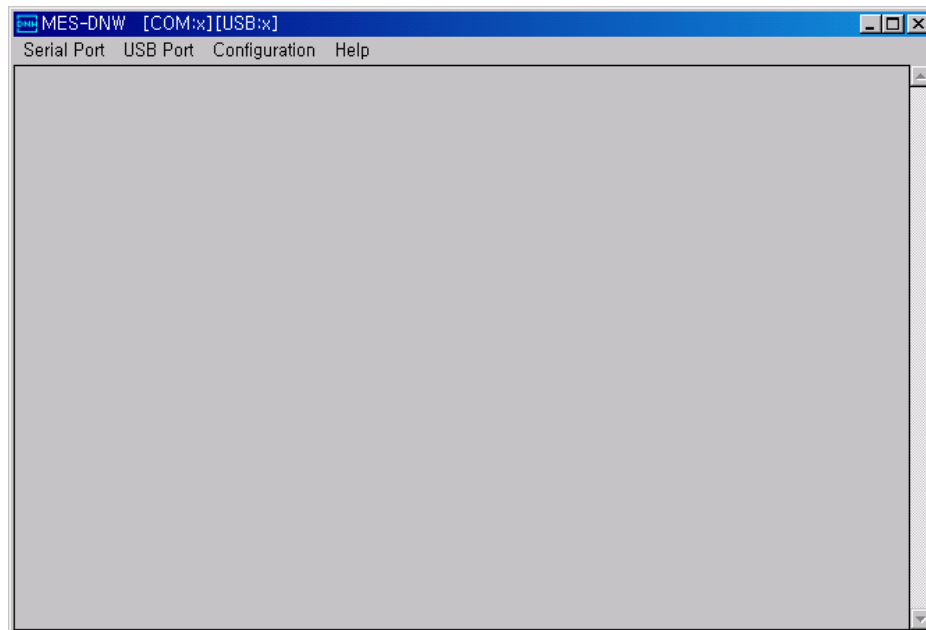
- Function as the second bootloader
- Function as a USB downloader (can download OS image file from PC to SDRAM(or NAND flash) through USB 1.1 device port)
- Function as a diagnostic test program
- Function as a Logo graphic file loader

1.3. The Preparations for using the MagicLoader

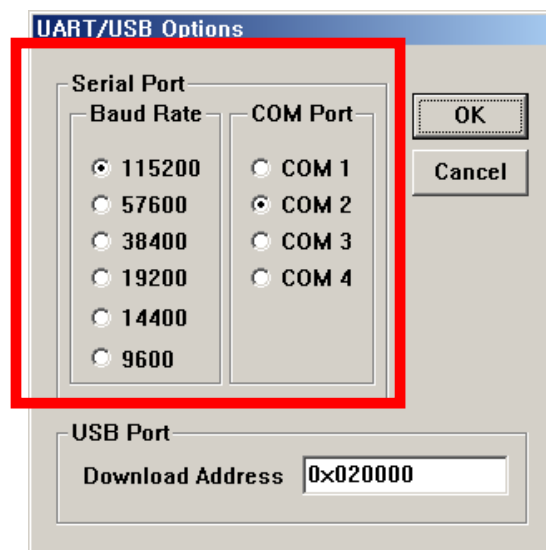
- Compile Tool : ARM Developer Suite V1.2
- USB cable
- UART cable
- DNW Terminal program
- USB driver for DNW

2. The Usage of the MagicLoader

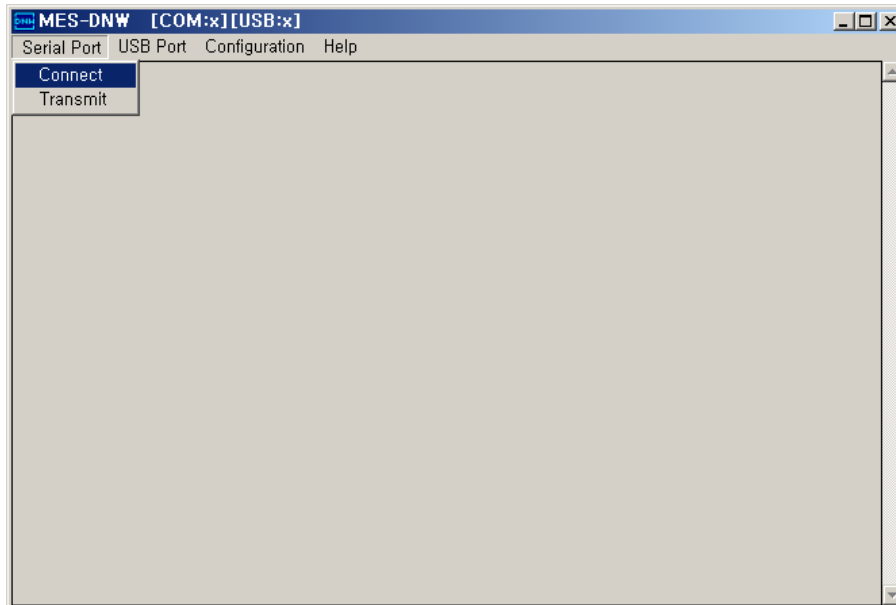
- (1) Connect the Serial Cable(UART Cable) between PC and KIT
- (2) Execute the DNW Terminal



- (3) Select **Configuration** → **Option** and set the COM port of PC and Baud rate

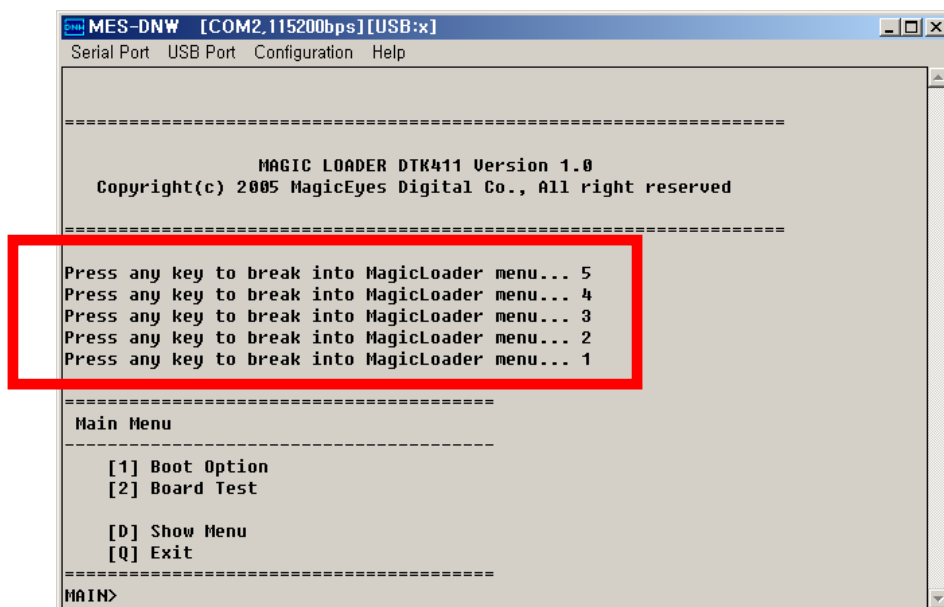


- (4) Select **Serial Port** → **Connect**



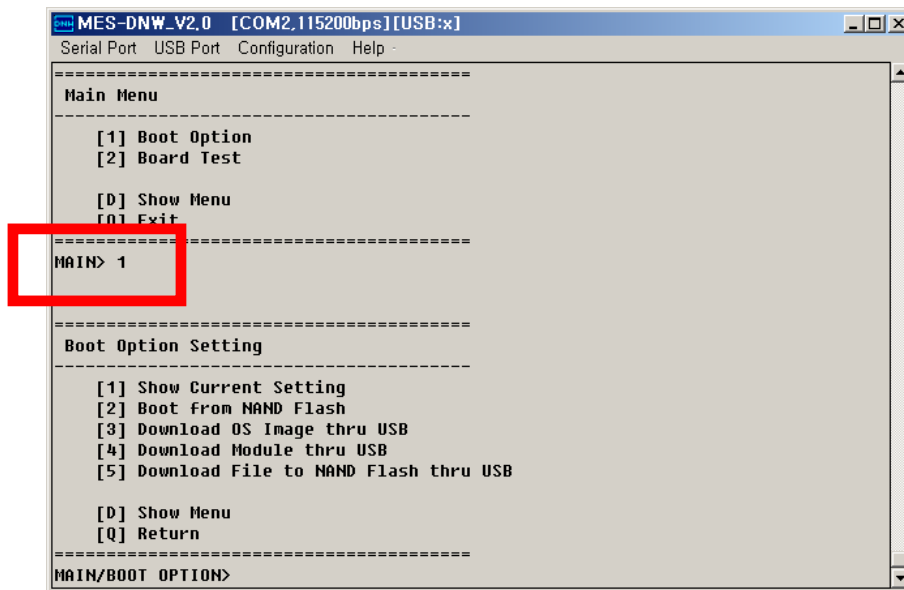
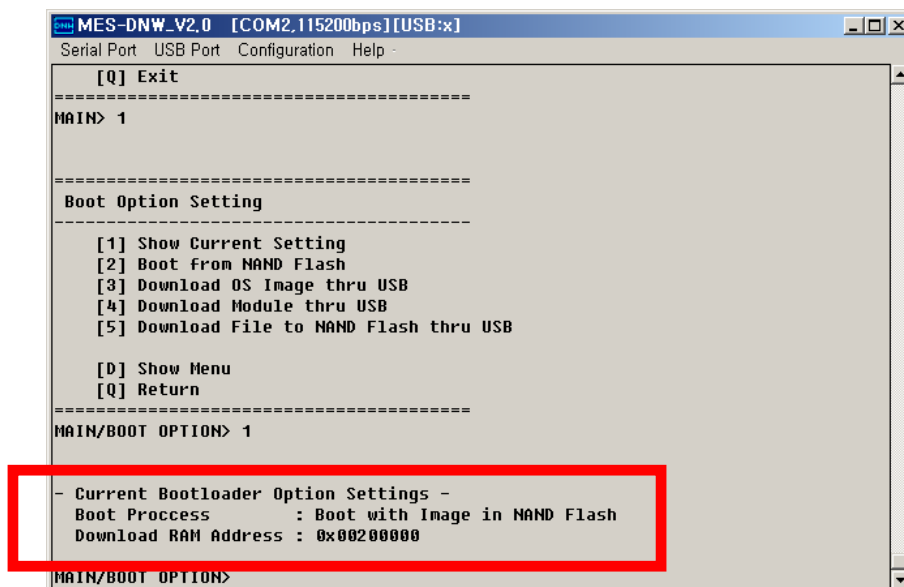
- (5) Turn on the KIT Power

- (6) If the MagicLoader starts, you can see the following message, 'Press any key to break into MagicLoader menu...' for 5 seconds. To change the Menu of MagicLoader, press any key of Keyboard. If there is no user's input for 5 seconds, MagicLoader will run according to the setting value which is stored finally.

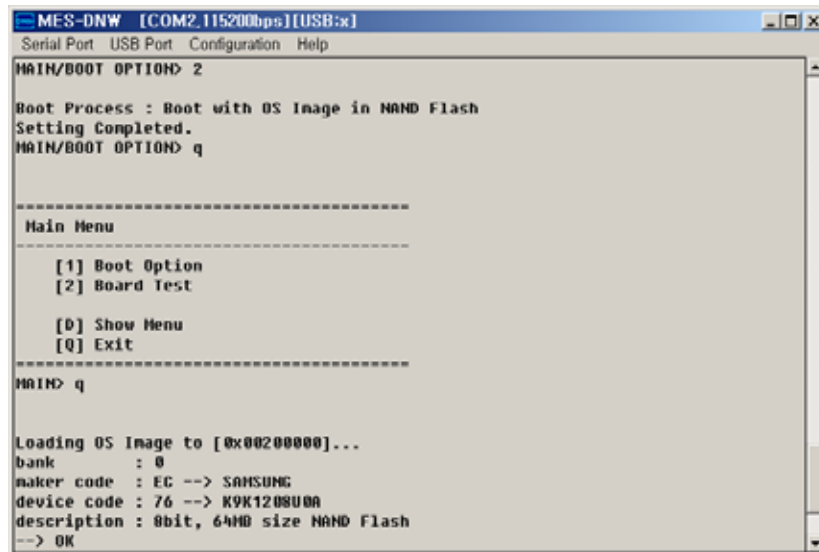


(7) Description of **Main Menu**

- [1] **Boot Option** : User can select Boot Option among the **Boot Option Setting** menus.
- [2] **Board Test** : User can test several functions of KIT.
- [D] **Show Menu** : User can see the Menu again.
- [Q] **Exit** : Escape from **Main Menu** settings.

(8) To enter the **Boot Option**, type 1 and press Enter key of Keyboard(9) [1] **Show Current Setting** : User can see the current setting value of Boot Option

- (10) [2] **Boot from NAND Flash** : If user set this option, KIT will be boot with OS image in NAND Flash



```

MES-DNW [COM2,115200bps][USB:x]
Serial Port  USB Port  Configuration  Help

MAIN/BOOT OPTION> 2

Boot Process : Boot with OS Image in NAND Flash
Setting Completed.
MAIN/BOOT OPTION> q

=====
Main Menu
=====
[1] Boot Option
[2] Board Test

[0] Show Menu
[Q] Exit
=====
MAIN> q

Loading OS Image to [0x00200000]...
bank      : 0
maker code : EC --> SAMSUNG
device code : 76 --> K9K1208U0A
description : 8bit, 64MB size NAND Flash
--> OK
    
```

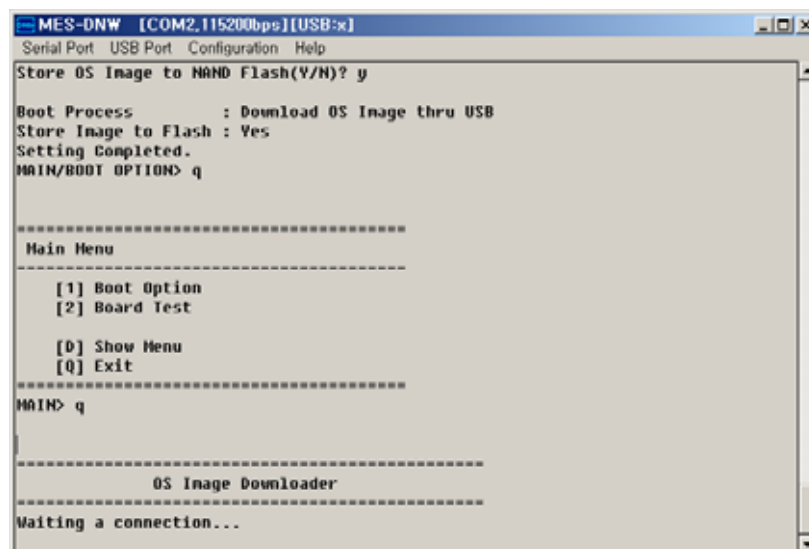
Type q and press Enter key to escape from **Boot Option Setting**

Type q and press Enter key to escape from **Main Menu**

- (11) [3] **Download OS Image thru USB** : User can download OS image file through USB1.1 device port. There are two options.

Y : Store OS image file to NAND flash

N : Do not store OS image file to NAND flash



```

MES-DNW [COM2,115200bps][USB:x]
Serial Port  USB Port  Configuration  Help

Store OS Image to NAND Flash(Y/N)? y

Boot Process      : Download OS Image thru USB
Store Image to Flash : Yes
Setting Completed.
MAIN/BOOT OPTION> q

=====
Main Menu
=====
[1] Boot Option
[2] Board Test

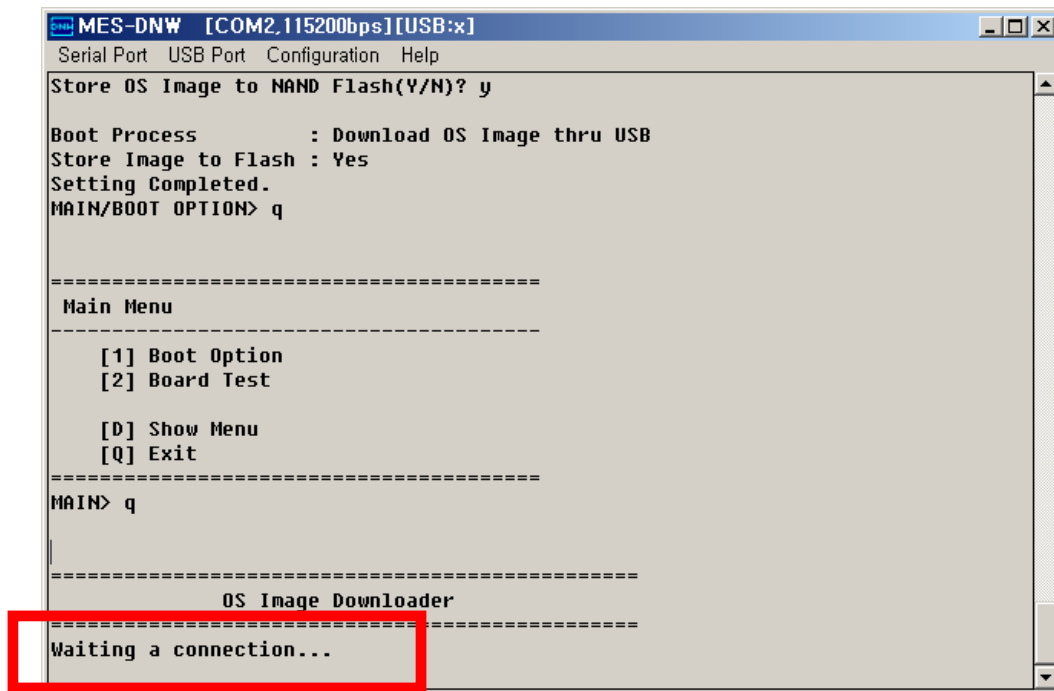
[0] Show Menu
[Q] Exit
=====
MAIN> q

=====
OS Image Downloader
=====
Waiting a connection...
    
```

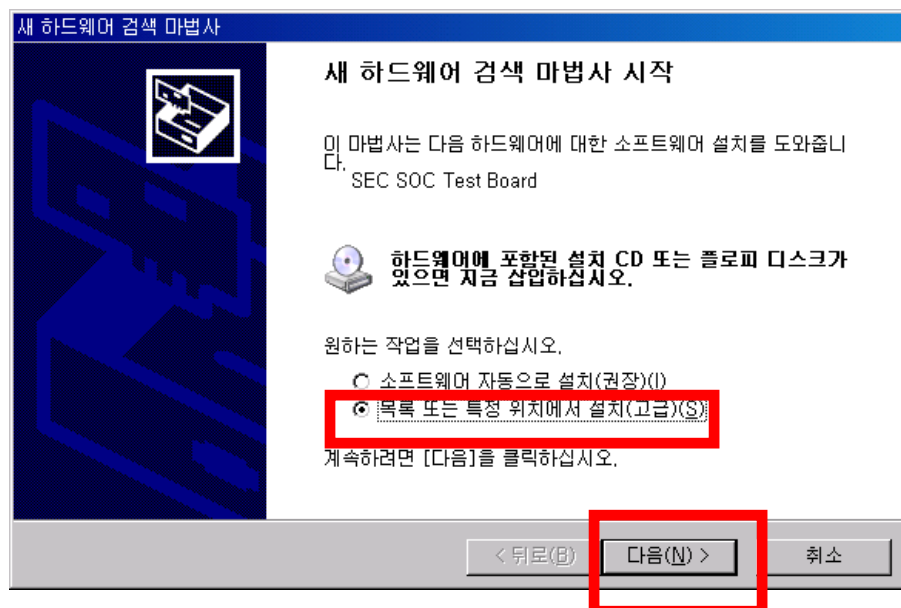
Type q and press Enter key to escape from **Boot Option Setting**

Type q and press Enter key to escape from **Main Menu**

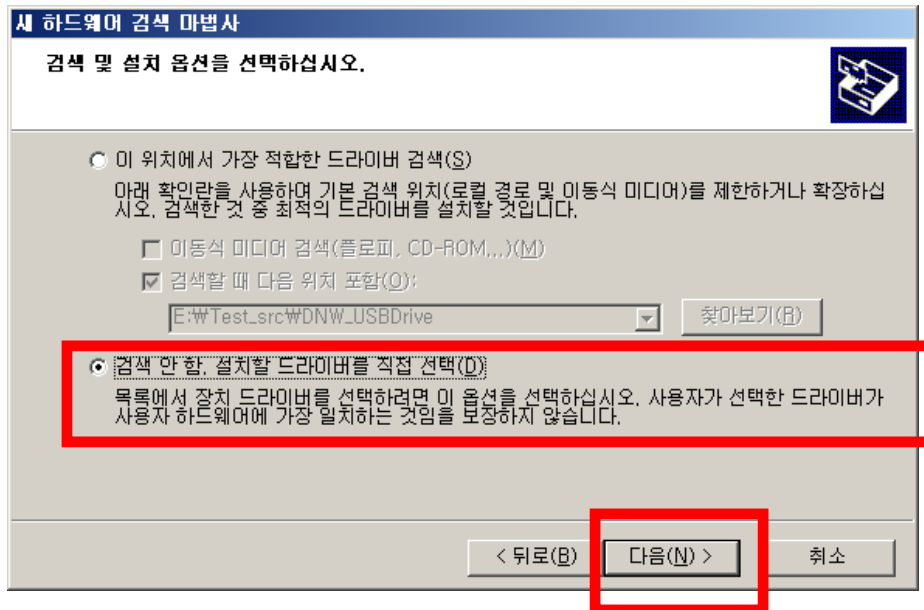
- (11-1) 'Waiting a connection...' message will be printed in DNW terminal. Connect the USB cable between PC and KIT.



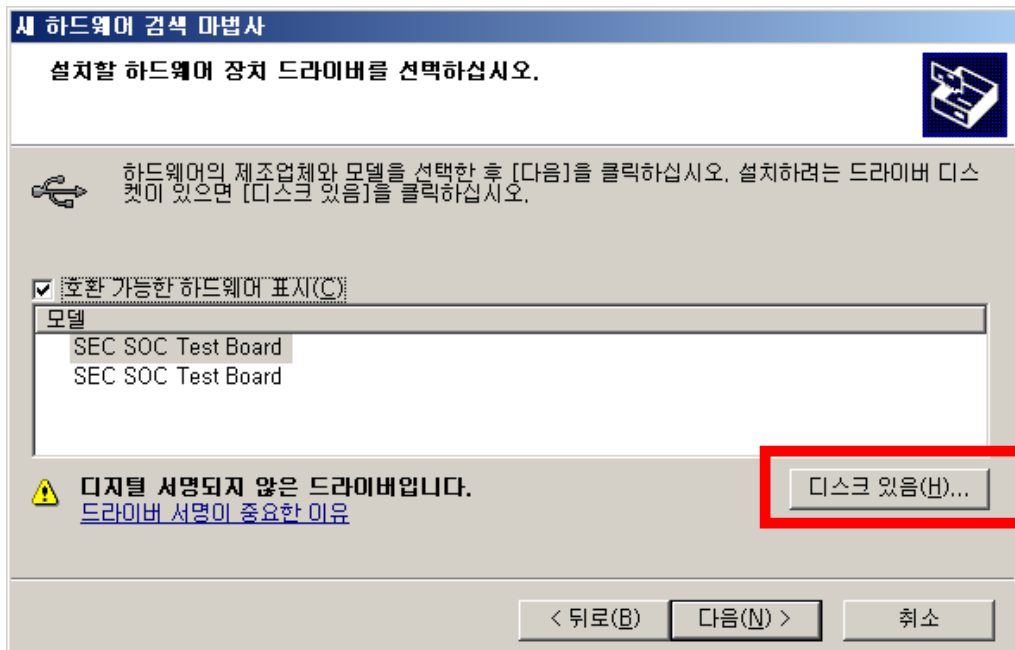
- (11-2) PC will detect new hardware and 'New Hardware Searching Wizard' window will be popped up. There will be a two options. Do not select 'auto installation' option, select 'installation from list' option and click 'Next' button.



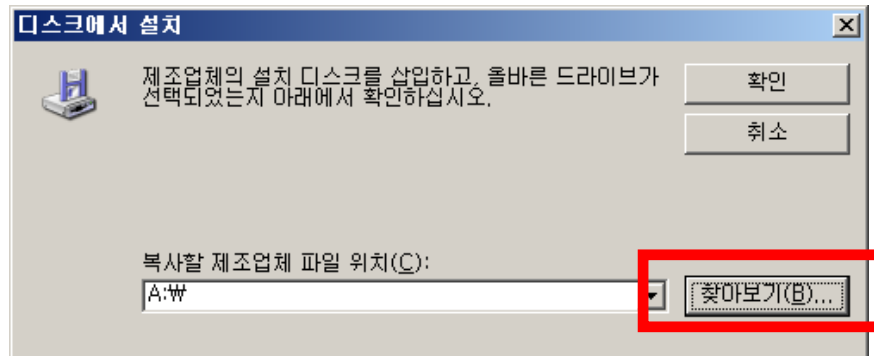
(11-3) Select 'Not search. Select driver directly' option and click 'Next' button



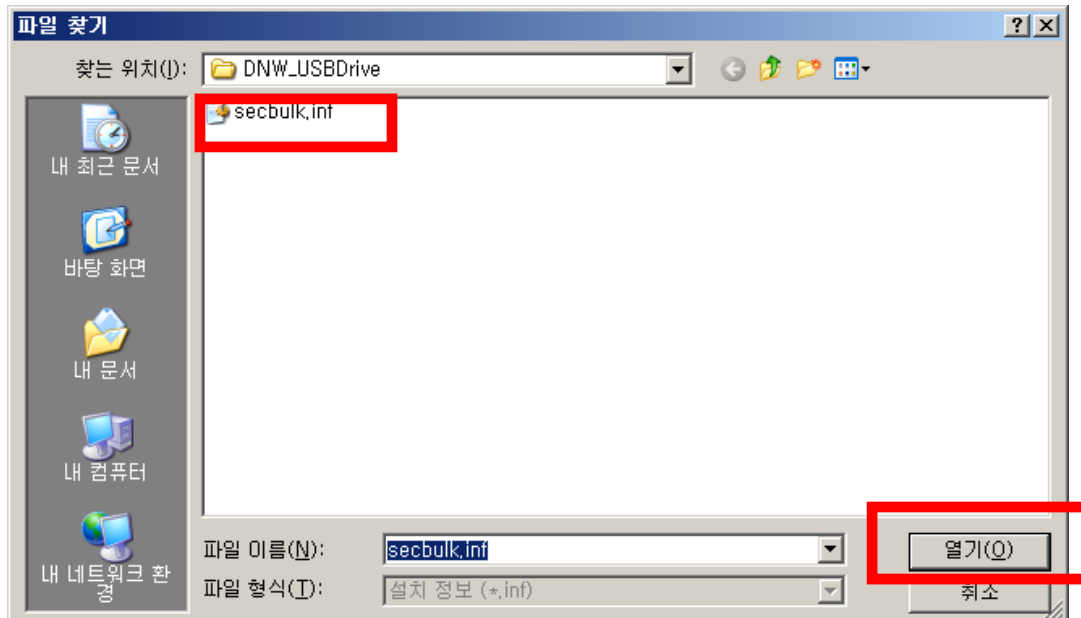
(11-4) Click 'Disk' button



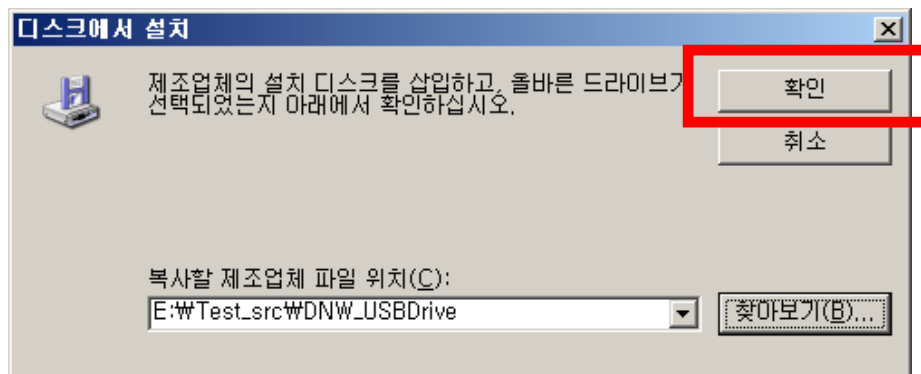
(11-5) Click 'Search' button.



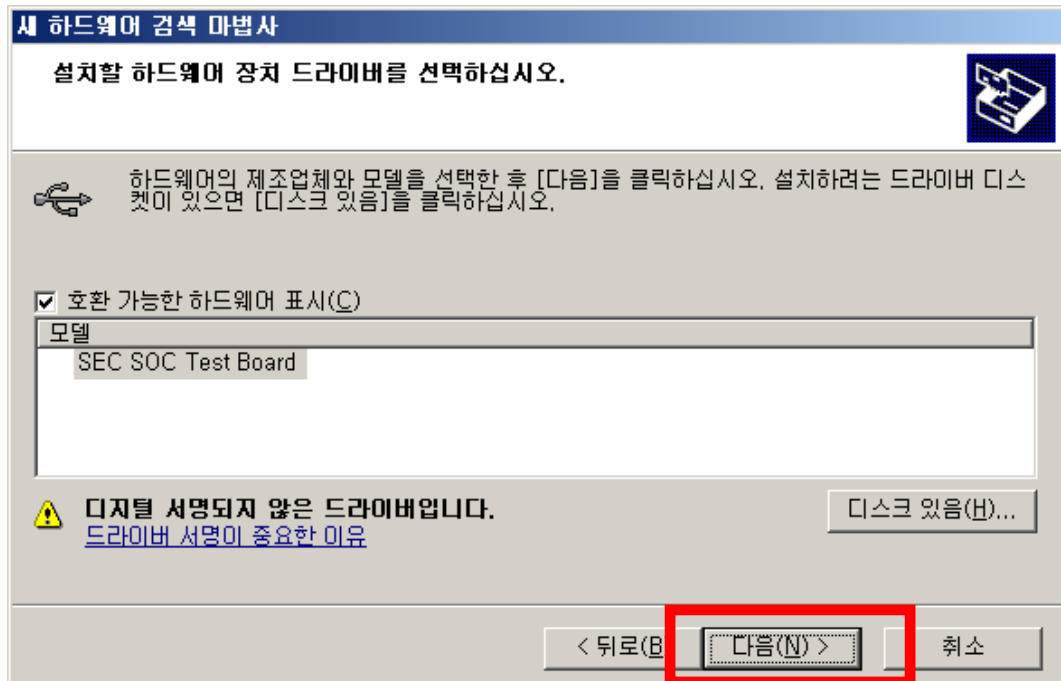
(11-6) Select 'secbulk.inf' file and open it.



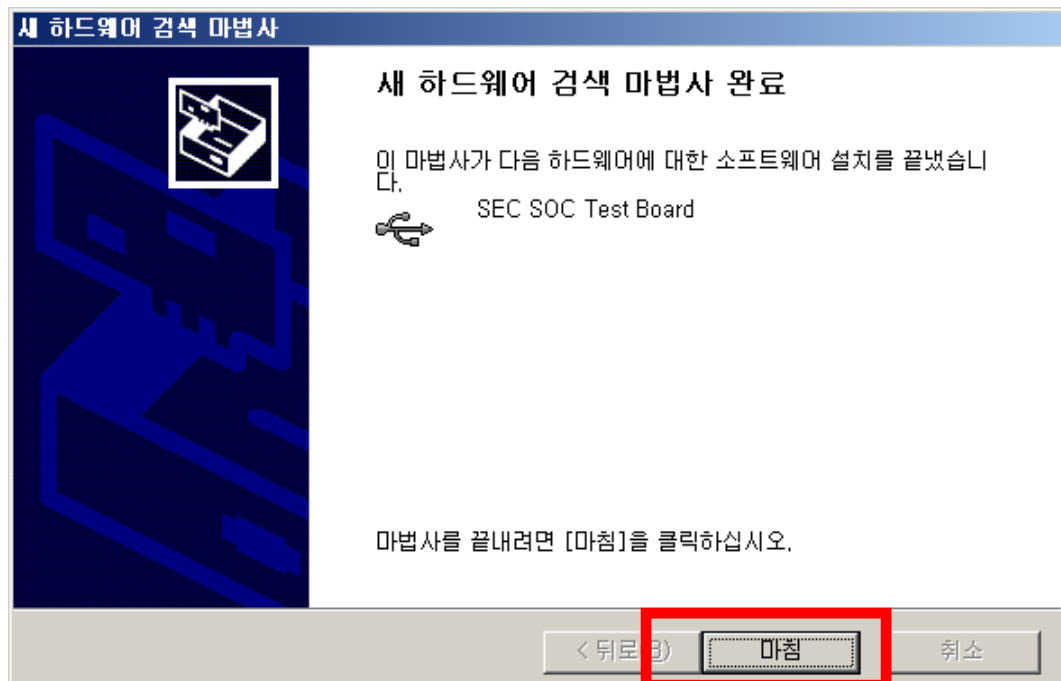
(11-7) Click 'Confirm'(or OK) button



(11-8) Click 'Next' button



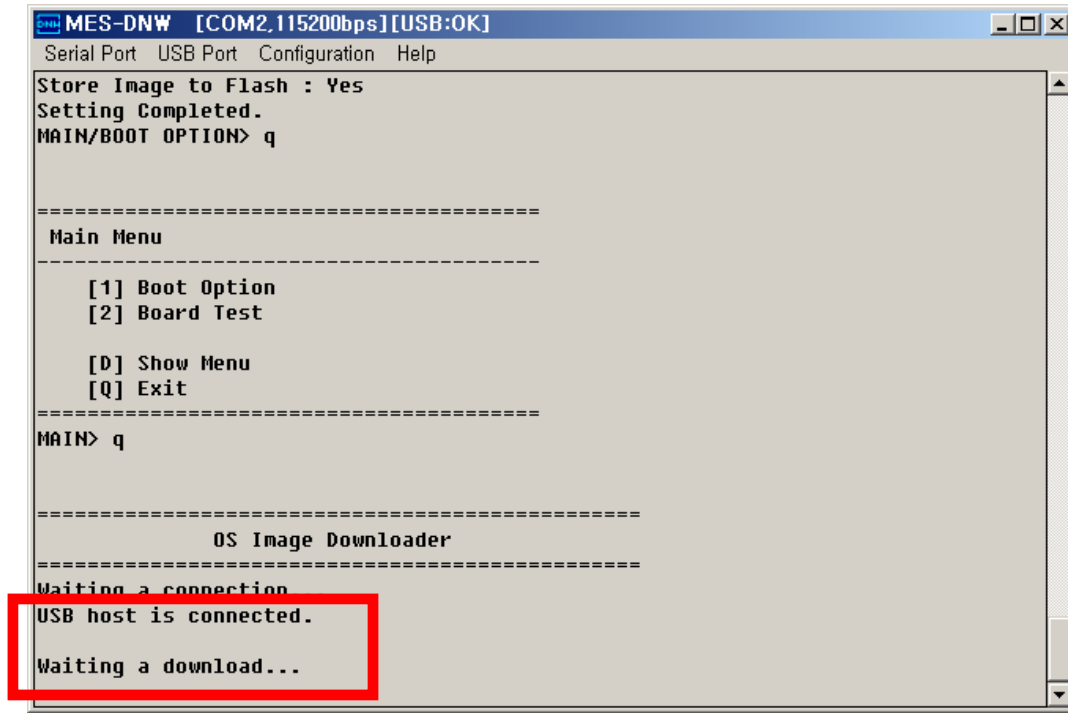
(11-9) Click 'Finish' button



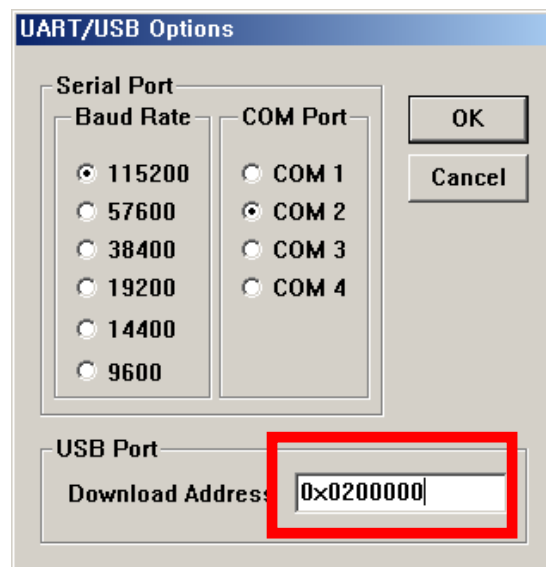
(11-10) If the driver installation is finished successfully, you can see the below message :

'USB host is connected.'

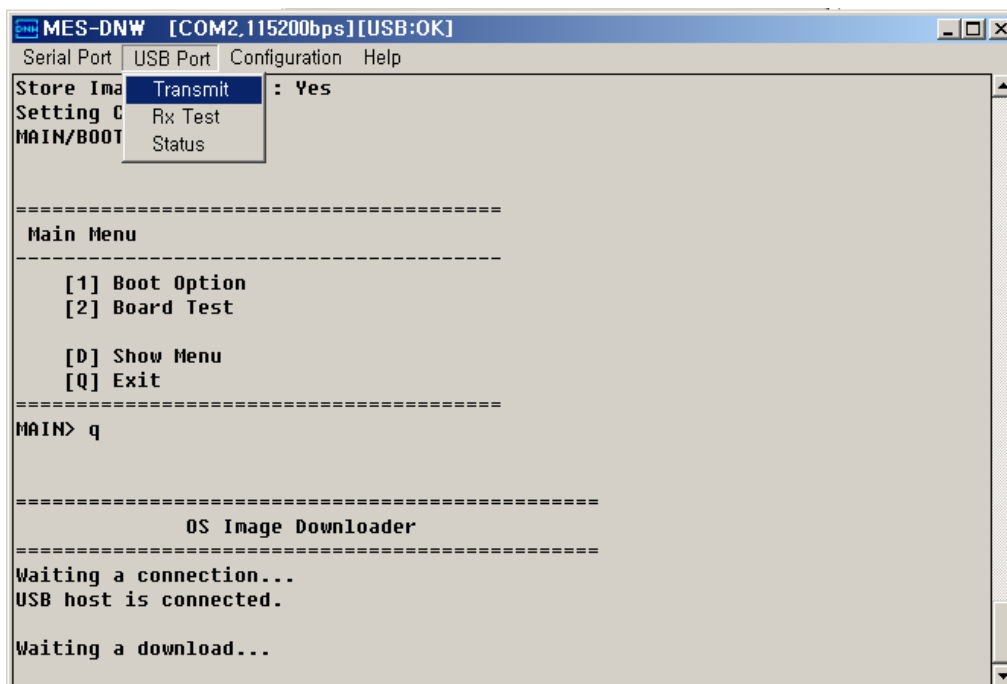
'Waiting a download...'



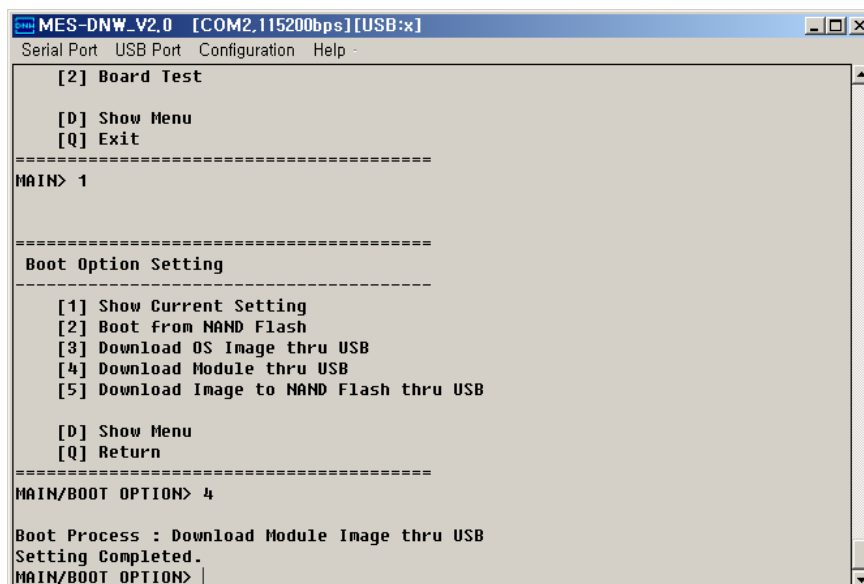
(11-11) Click **Configuration** → **Option** and set **Download Address** to 0x200000.



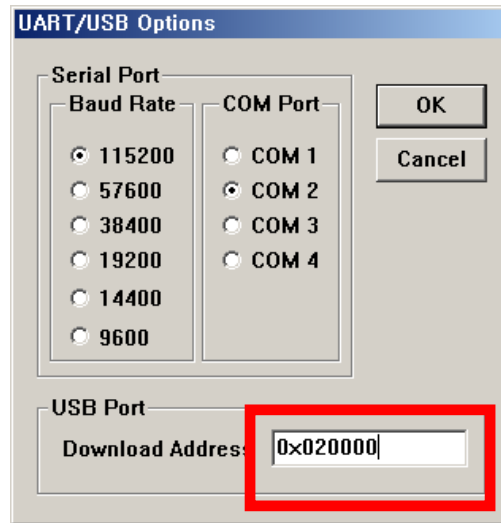
(11-12) Click **USB Port** → **Transmit** and select the OS Image file to download.



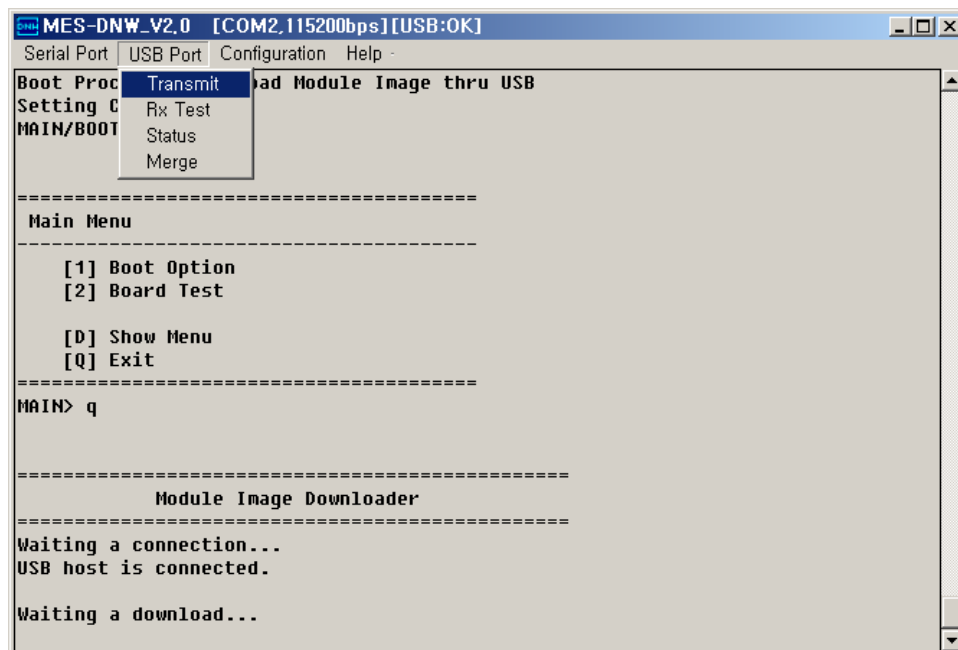
(12) [4] **Download Module Image thru USB** : User can download project image file(ex. Magicloader image file) from PC to SDRAM.



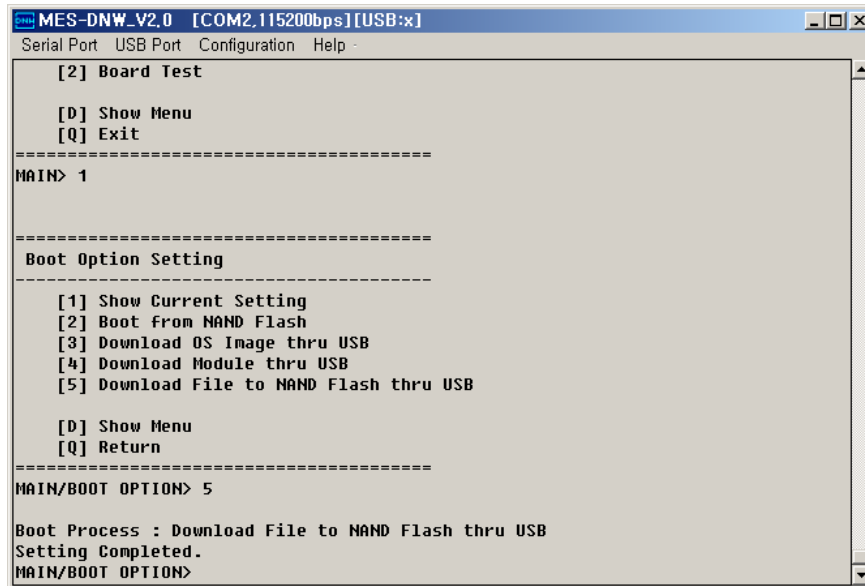
- (12-1) Click **Configuration** → **Option** and set the **Download Address** to 0x20000



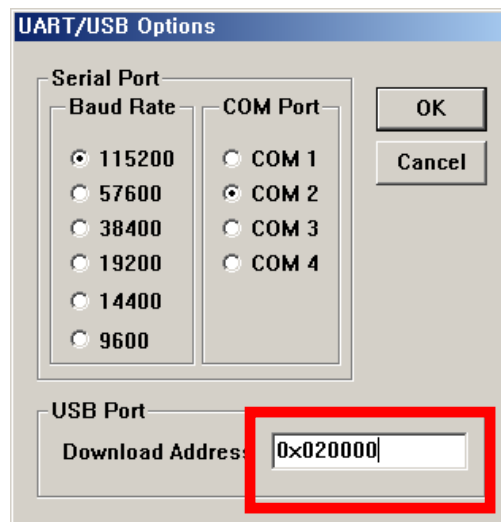
- (12-2) Click **USB Port** → **Transmit** and select Project Image file to download.



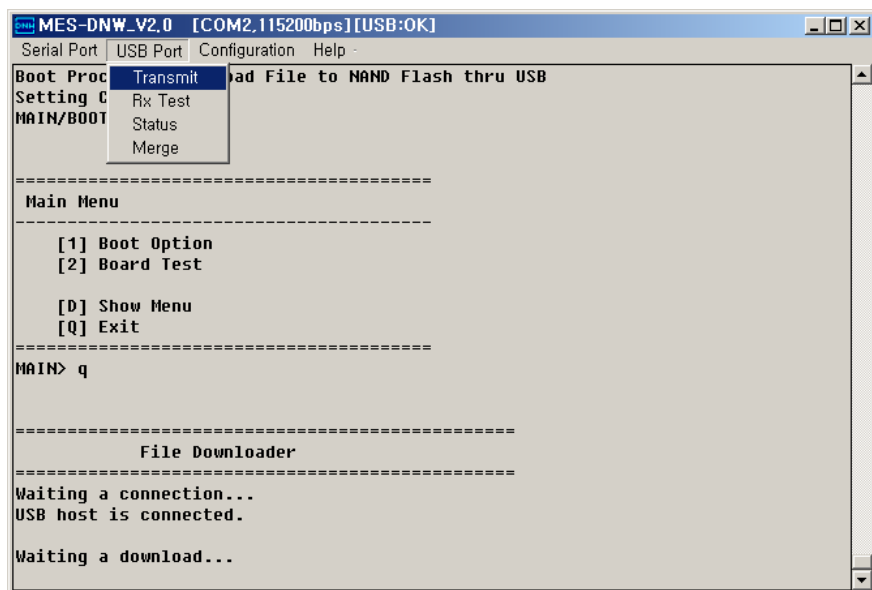
- (13) [5] **Download file to NAND Flash thru USB** : User can download any kind of file from PC to NAND Flash



- (13-1) Click **Configuration** → **Option** and set the **Download Address** of NAND Flash



(13-2) Click **USB Port** → **Transmit** and select Image file to download.

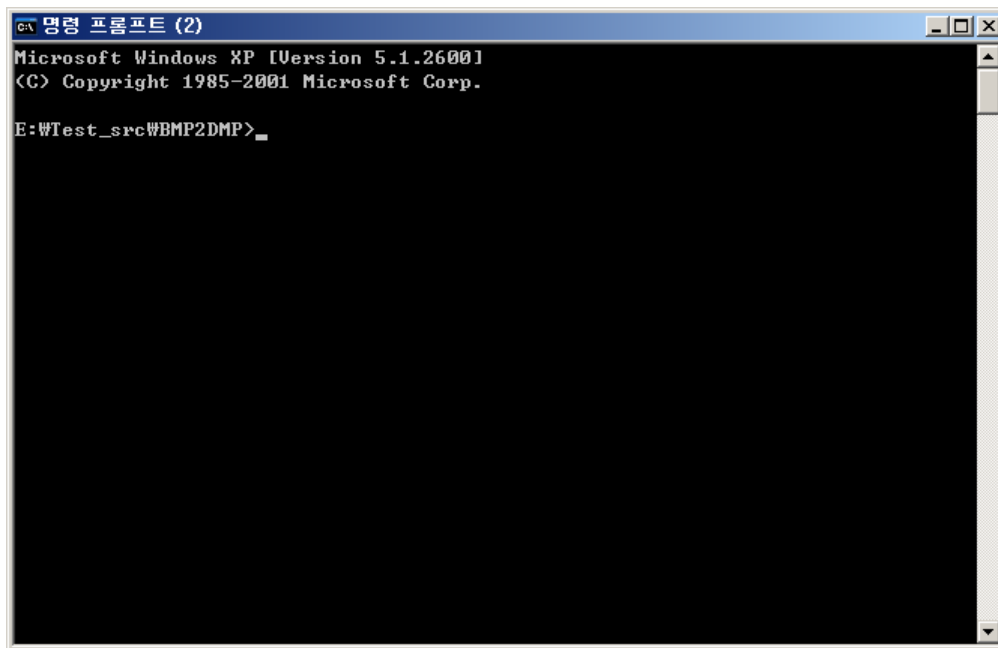


3. How to display logo graphic file

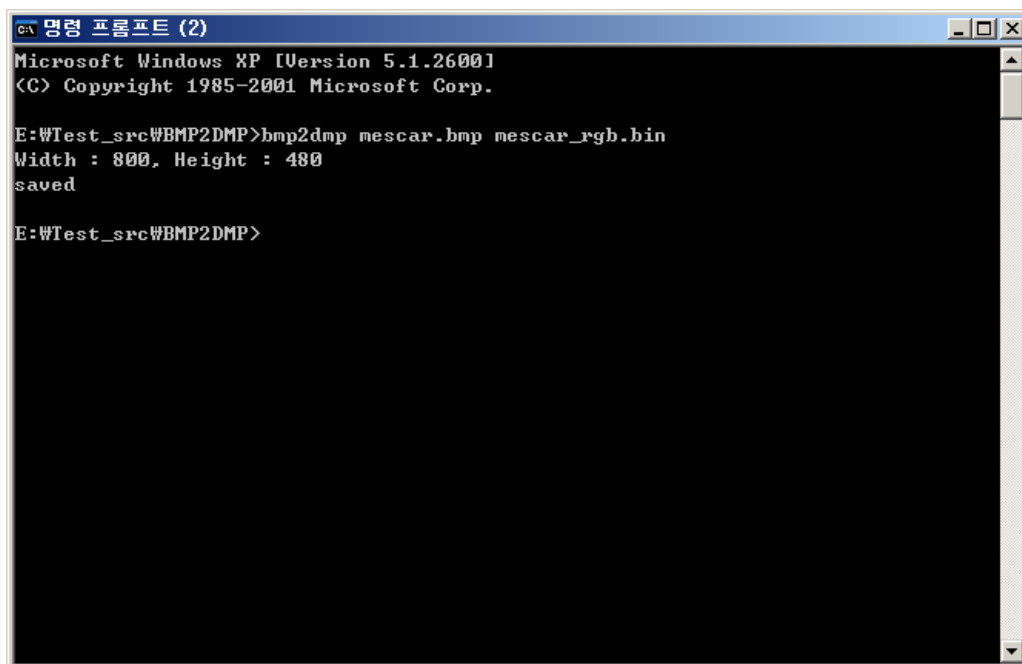
3.1. Make logo graphic file

- Logo graphic file must be 16bit RGB DATA Format
- BMP2DMP.exe is a file format converter from 24BIT BMP file to 16bit RGB DATA Format

(1) Command execute

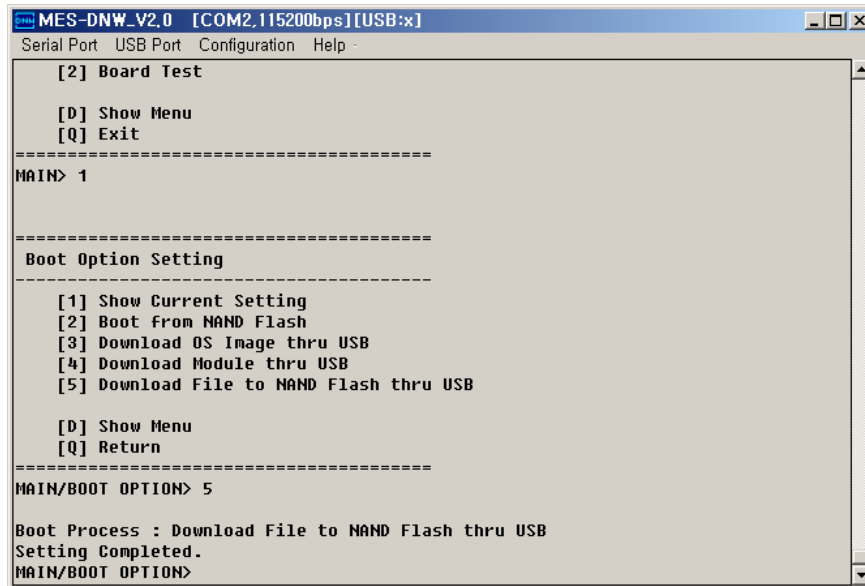


(2) BMP2DMP.exe 24BIT BMP filename dstfilename (ex:24bit 800x480 BMP file)

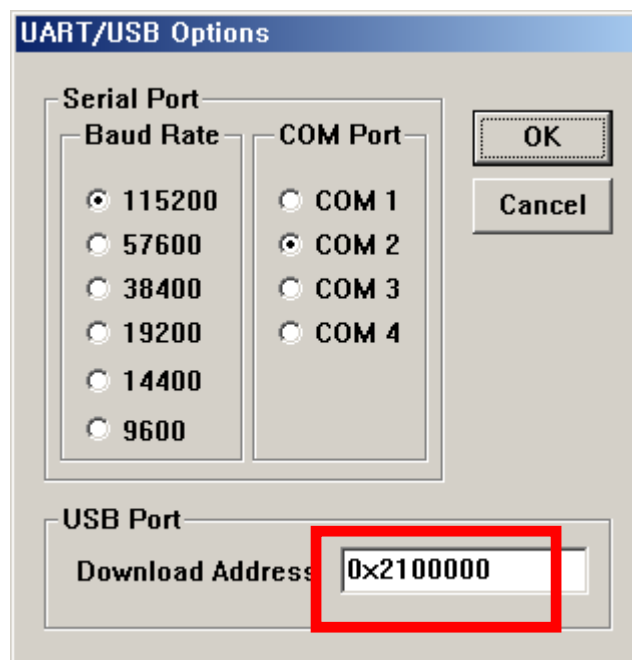


3.2. Write logo graphic file to NAND Flash

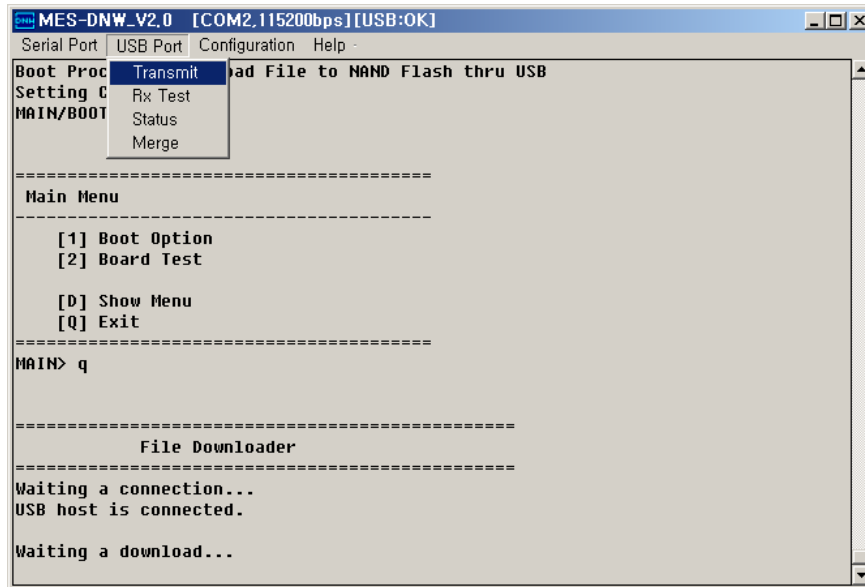
- (3) Select '[5] Download file to NAND Flash thru USB' of Boot option Setting menu



- (4) Click **Configuration** → **Option** and set the **Logo Address** of NAND Flash (0x2100000)



- (5) Click **USB Port** → **Transmit** and select logo graphic file to download.



3.3. Display Device Configuration

- (1) Default Display Configuration Value is for Hitachi 7" Wide VGA LCD of DTK411
- (2) User can modify the Display Configuration Value for user's LCD. To do this, open the 'OEM_MMSP2_SYSTEM_DTK411.h' file and add or modify the LCD Display Configuration Value, like as Horizontal width, Vertical width, clock timing, and so on